Ch. 9 — P01 (A) Distribution Fitting by Hand

Problem Bank | Chapter 9 | Type: A | Difficulty:

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You have collected 20 inter-arrival times (in minutes) from a university help desk:

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2.3, 8.1, 0.4, 5.6, 12.4, 1.7, 3.9, 0.8, 6.2, 15.3,
0.2, 4.4, 9.7, 2.1, 0.6, 7.8, 3.3, 1.1, 5.0, 11.2
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Tasks

1. Compute: n, mean $\bar{x}$, sample variance s^2, CV $= s/\bar{x}$, skewness. Based on CV alone, hypothesize whether the data is consistent with Exponential, Gamma (shape > 1), or heavy-tailed (CV > 1).
2. **Exponential MLE:** The MLE for the Exponential rate is $\hat{\lambda} = 1/\bar{x}$. Compute $\hat{\lambda}$ and the fitted mean.
3. **Gamma MLE** (method of moments as an approximation): Using $\hat{k} = \bar{x}^2/s^2$ and $\hat{\theta} = s^2/\bar{x}$, compute the method-of-moments estimates for shape k and scale θ. What is the fitted mean and variance?
4. **KS test by hand:** Compute the empirical CDF $F_n(x_i) = i/n$ for $i = 1, \ldots, n$ after sorting the data. For the Exponential fit, compute $F(x_i; \hat{\lambda}) = 1 - e^{-\hat{\lambda} x_i}$. Find $D_n = \max_i |F_n(x_i) - F(x_i)|$.

   The critical value for a two-sided KS test at $\alpha = 0.05$ with $n = 20$ is approximately 0.294. Do you reject the Exponential hypothesis?
5. Which distribution do you select and why? Write a one-paragraph input model specification for this dataset.